NightSweep 1.8 Alpha build 2 — All Functions Reference

Technician README · Updated November 12, 2026

This reference explains the working functions in NightSweep 1.8 Alpha build 2: where to find them, what to enter, how to interpret the result, and which bundled example lets you practice. For the current 2.0 Alpha build 3 source, it covers 35 task screens and 53 entries in the Practice library's feature index.

2.0 Alpha build 3 (Carpe Noctum, in development) includes Field calculators, Limits and sources, the DOCSIS health card, Leakage, Station worksheets, Outage first response, Night log and shift handoff, Personal sweep reminders, Attached sweep grading, Impairments, Fiber, High split, Why cards, Proof of performance, Study mode and Supply PM. The seven exercise lengths are complete 22, six-active 16, custom curves 21, OEM AGC 8, DSIM 9, manual/thermal 7 and GS7K 10. The coverage index lists all 53 feature choices. Safety was removed by Spencer's decision on November 12, 2026; legacy saved data is left untouched.

For installation, use the separate Android and iOS installation guide. This reference applies to the shared Android/browser application. Browser-only offline and update controls appear on Home; Android's native save/share/print dialogs appear when using the APK.

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Working rules and navigation

Start a real job with **Job → Details → + New blank job**. Identify the actual source output, enter launch readings, build the route, then use the calculations, setup guides and records you need. A single-span calculation does not require completion of the advanced acceptance forms.

Main work area	Task screens	Purpose
Home	Home	Current job, task shortcuts, practice, backup and browser offline status
Route	Setup; Cable route; Amp gain; Amp / node setup; Catalog; Station	Starting signal, RF topology, active gain and exact-model references
Test	Meter setup; TDR; Predicted levels; Frequency; Readings; References; Calculators; Limits; DOCSIS health; Leakage; Impairments; Fiber; High split; Why cards; Proof of performance; Study	Guided test records, predicted response, distance and measurement comparisons, quick calculations and the source of every limit
Power	Load; What-if; Supply PM	Independently entered power circuit, an added-load trial, and a personal supply notebook with span voltage drop
Job	Details; ONX files; Checklists; Outage; Acceptance; Run sheet; Night log; Sweep due	Local jobs, original evidence, work notes, acceptance records and reports
Header / Home	Help & methods	Task help, methods, limits and build self-check

The small task tabs choose a function within the main area. The point selector and **Previous/Next** choose the route location; they do not change its feed parent. **More at this point** exposes tap faces, drop, map notes and arithmetic.

Keep these distinctions while using every tool:

- **Prediction** is calculated from entered data. A meter reading is a separate entry; neither automatically becomes acceptance.
- **Manufacturer capability**, local design target, installed configuration and available spectrum are separate fields.
- A dash or blank result means unavailable, missing, invalid or outside supported data. It is not zero. A deliberately entered zero is different.
- Use the same frequency, measurement plane and signal-power convention when comparing readings. Apply only test-point correction the meter has not already applied.
- **Vecima nodes / AOI actives:** local 204/258 MHz workflow, forward Sweepless Sweep and XPERTrak return sweep. **GS7K, GainMaker and the other fleet families:** local 42/54 MHz workflow, forward and Return Sweepless Sweep.
- Non-high-split design anchors are **31 dBmV at 54 MHz / 43 dBmV at 860 MHz**. The separate **38 dBmV return TX reference** does not specify a receiver target, injection setting, channel or measurement plane.

Header and Home controls

Where: the header and Home. **Practice:** Complete field workflow; Custom curves for Help and methods.

Function	Enter or do	Result and limit
Main and task tabs	Choose a work area, then its task	Opens the relevant controls while preserving the current job
Files → Back up project	Export the working project and finish the device's save action	Portable JSON with supported job data and original evidence; verify the file outside the app
Files → Open project	Choose a NightSweep project JSON	Validates the incoming project before loading; export important current work before replacement
Files → Export RF levels	Export the current route table	Predictions and context in CSV; unavailable results remain blank
Files → Print report	Open the native or browser print dialog	Printing/PDF depends on the device and print service; use Run sheet for a focused field summary
Help & methods	Open help from the header or Home	Task guidance, calculation definitions and limits
Palette button (header: Switch to Sun palette or Switch to Night palette)	Select the display palette	Changes presentation, without changing inputs or calculations
Working-job card	Read the job name, asset and sample/training warning	Helps prevent entering real readings into a sample job
Task shortcuts	Choose route, TDR, gain, readings, power, sweep setup or run sheet	Opens that live task using the current job
Home statistics	Review farthest route distance and active count	Summarizes the entered RF topology; an active count alone does not prove a serial cascade
New-user quick start	Expand the short workflow	Opens existing task controls; it does not complete them automatically
Home backup / project opener	Use the Save/Open project shortcuts	Same full-project workflow as Files; not automatic cloud synchronization
Status and warning messages	Read the nearby message after an action fails	Identifies missing inputs, storage/file errors or a blocked operation; preserve backups before repairing storage

Browser Home also provides **Install app**, **Check for updates**, **Install update**, **Reload app**, **Retry offline setup** and **Protect local storage** when supported. Their behavior is explained under Offline use, updates and Help.

Practice library and example selection

Where: Home → Practice library. Choose an exercise, or choose a feature to load its seeded example and open the matching guided step. These examples are bundled; no manufacturer website is needed to load them.

Exercise name	What is loaded	Guide / supplied evidence
Complete field workflow	GS7K source, two GainMaker GaN HGD actives, Scientific Atlanta / GigaXtend / Technetix passives, branched 42/54 route, power, readings and records	22 steps; five synthetic CSV originals and one original-only SVG diagram
Six active devices in series	Vecima EN9000 source and six serial AOI actives on 204/258	16 steps; no supplied meter captures
Custom curves and reverse TDR	Manual launch table, custom cable/passive tables, exact-frequency entries, reverse TDR, field calculators, limit sources, leakage, outage first response, impairment cards, the fiber color finder, the high-split reading aid, the why cards, the FCC proof-of-performance card, study mode and the supply PM notebook	21 steps; no supplied meter captures
GainMaker OEM AGC	Separate 42/54 conventional OEM detector/pilot path	8 steps; no supplied meter captures
GainMaker / DSIM	Separate 42/54 fitted DSIM identification, historical code lookup and incomplete verification	9 steps; no supplied meter captures
GainMaker manual / thermal	Separate 42/54 manual/thermal installed-control path	7 steps; no supplied meter captures
GS7K conventional node	Standard GS7000 1.2 GHz / 42/54 setup starter; actual launch/readings/correction unverified	10 steps; no RF route or supplied meter captures

Opening or resetting training preserves the working job locally before switching. A full or unreadable named-job list, an evidence operation in progress, or a preservation failure stops the switch. Read the message and keep the existing job; do not clear storage to force it. A local retained copy is not an external backup.

Use **Open step** to reveal the relevant controls, try them, then mark **Reviewed**. **Practice guide** in the training banner returns to the guide. Progress is saved with that practice copy; it does not mark field inspections or acceptance complete. Reset preserves the edited copy before restoring the exercise baseline.

Older practice copies remain readable. If their equipment no longer matches the current complete exercise, the app hides mismatched guidance and offers an explicit reset. Renaming, editing, saving or exporting a practice job does not remove its synthetic identity.

The complete workflow uses GS7K/GainMaker GaN, not the AOI six-active path. Its GS7K hardware capability is 1218 MHz, but the entered operating/planning upper is **1000 MHz** and live spectrum **860 MHz**. The selected GaN active covers 54–1002 MHz and the Scientific Atlanta route passive stops at 1000 MHz. Do not infer end-to-end 1218 MHz operation from the source model.

Job details and named jobs

Where: Job → Details. **Practice:** Complete → Job details; any exercise's backup step.

Function	Inputs and use	Output and limit
Job identity	Name, node/asset, date and lead technician	Current-job context used in reports and exports
Work-order fields	Ticket, address/location note, symptom, start/end times, weather note, resolution and parts used	Stored with the job; a time-order warning preserves the entered text and does not change calculations
Linked metadata	Edit work-order number, node or lead in Details or Records	Keeps those shared identity fields consistent; other record fields remain their own entries
Node details	Identify model, installed band, asset, RF port/leg, modules, operating upper and notes	Describes source identity; model selection does not supply measured launch levels
+ New blank job	Start a clean field job after protecting current work	Removes sample inputs from the new working job; check its name before field entry
Save now	Save the desired working version to the device's job library	Local named copy; save/export important revisions deliberately
Open	Select a stored job	Restores that job's data and attachment references; storage failure blocks unsafe switching
Duplicate	Make a separately named copy	Uses the latest working state when duplicating the active named job; updated original and new copy are written together. A failed write leaves the library unchanged; training stays synthetic
Delete	Remove the intended named entry after exporting anything needed	Not a recovery workflow; check the job identity before confirming

Autosave and named jobs use this app/browser's storage. APK, Chrome, another browser/profile, another phone and a Home Screen web app may use different stores. No account or automatic job synchronization is provided. Use one NightSweep window at a time and transfer jobs with verified full-project exports.

RF setup and launch source

Where: Route → Setup. **Practice:** Complete → Launch; GS7K conventional node → Identity; Custom curves → Launch.

Function	Inputs and use	Result and limit
Launch-source choice	Select **Node output**, **Amplifier / LE output**, or **Manual / other output**	Identifies the physical starting output; it does not add an artificial gain block or fill levels from a rating
Node / active identity and output	Match exact model, installed split/module and selected RF port; use the job node ID shortcut where appropriate	Makes the source/manual context explicit; source changes preserve readings for review rather than making them valid for the new connection
Source setup & manual	Open the selected equipment's reference	Requires the matching installed variant and revision; external originals need internet
Low/high launch markers	Enter two frequencies and corresponding levels at the same source plane	Defines the two-point source curve; ordered, valid in-band markers are required
Launch test point	Mark raw TP readings and enter only remaining coupling	Adds the stated correction once; use zero/off when the meter already supplies mainline-equivalent levels
Launch curve	Choose square-root frequency, linear MHz or **My level table**	Uses the selected interpolation model; it is not a measured EQ response
My level table	Enter one `MHz, dBmV` pair per line	Uses actual entered source points; stops outside the first/last supported point
Extend two-point launch slope	Enable only for an intentional estimate outside the two anchors	Extends the two-point launch estimate; it does not extend cable/passive table coverage or establish live carriers
Available spectrum ends	Enter the known live-spectrum upper limit	Distinguishes live from planning predictions; hardware capability is separate
Passive loss basis	Choose **Nominal / typical** or **Maximum**	Nominal can use a labelled maximum fallback when no nominal curve exists; read the stated basis
Cable temperature	Enter RF cable temperature in °F	Adjusts supported built-in attenuation references; custom cable values are already at actual temperature
Next: cable route	Continue to the route editor	Starts at the chosen source output without inventing another station

In Complete, **11/23 dBmV +20 dB** at **54/860 MHz** produces **31/43 dBmV** launch. This is synthetic arithmetic, not proof those physical carriers or readings are present. The six-active AOI example instead uses 261/993 MHz and 34/46 dBmV.

Cable route, branches and point tools

Where: Route → Cable route. **Practice:** Complete → Route / Drop; Custom curves → Custom / Arithmetic; Six actives → Route.

Function	Inputs and use	Result and limit
Add, select and remove points	Add a cable span ending at equipment; select it with the point selector or Previous/Next	One point includes its incoming cable. Review child feeds after changing/removing a point; maximum 200 route points
+ Append after last tap	Use when the final route entry is a tap with a through output and a lower published face available	Appends through that tap, copying cable, incoming length, slack and custom cable data while choosing the next lower published face. It does not choose an arbitrary earlier branch or work after an ineligible final point
Point name and incoming feed	Name the location and choose the actual parent **output**	Builds the topology. A branch feeds from its selected splitter/tap port, not merely the same housing
Cable, length and slack	Select cable, enter mapped feet and additional slack	Adds both to attenuation/distance; do not add prefilled slack again
Apply this length to following spans	Confirm copying the selected point's incoming length	Applies to later entries in the route list, including later branches; not only the chosen electrical path. Review affected points and their separate slack
Equipment selection	Choose the exact family, type, part/face and terminal configuration	Uses the selected port curves; active, passive, cable-only and custom points have different behavior
Parent outputs and branch ports	Choose through, low-loss, high-loss, tap or other actual output	Preserves unequal splitter paths. Per-F-port tap output is not the through level
Cable-only point	Select no ending passive	Retains the cable span without adding invented equipment loss
Custom cable	Enter `MHz, dB per 100 ft` rows	Uses entered attenuation at actual temperature; no extra built-in temperature factor or out-of-range extrapolation
Custom passive	Enter `MHz, through dB, tap/branch dB` rows	Keeps the two output losses separate; missing data remains unavailable
Published faces	Compare same-family / same-port-count tap faces at the chosen frequency; choose a face to swap the part	Changes the catalog part and prediction; read through and tap loss separately
RG-6 drop off a tap port	Plan a drop from a selected tap output; enter feet, a custom cable table, then no splitter, a supported splitter or explicit custom loss	Calculates drop-end outputs independently of the mainline route; this is not a power-passing model

Function	Inputs and use	Result and limit
Remove drop	Remove the optional drop plan	Does not remove the mainline tap
As-built / map variance	Record a location-specific note	Saved with the point and shown in relevant reports; it is not a calculated cable measurement
Show the arithmetic	Expand the selected point's detailed calculation	Shows source/input, cable rate/temperature/feet, equipment loss or gain and output; unavailable steps stay visible as unavailable
Route cascade / distance summary	Follow the selected feed path and branch totals; scroll within the compact cascade in either direction, or focus its controls by keyboard	Shows separate path coordinates; all points remain reachable without making the phone page excessively tall. Branch footage is not added to the main endpoint
3-tap route example	Load the small RF-route sample after protecting work	A separate, simpler example; it is not the complete training job or the six-active run

Complete uses Scientific Atlanta **SAT MM 4-23**, GigaXtend **XSC-2-959** and **GTSG-TAP-4-20-STD**, and Technetix **OTTZ-4-17**. Its drop uses a **75 ft synthetic RG-6 table plus explicit custom 4 dB loss**, not an asserted premises-splitter model. Its main endpoint is 2380 ft from source; the branch cable-only endpoint is 2220 ft.

Amplifier gain and test-point helper

Where: Route → Amp gain. **Practice:** Complete → Gain; Six actives → Gain / Gain table.

Function	Inputs and use	Output and limit
Active-point selector	Choose an active already in the RF route	Edits that station; source identity alone does not create a route amplifier
Two measured/configured gain points	Enter the two frequencies and available operating gain at each	Calculates frequency-specific gain; match the installed configuration and measurement planes
My frequency/gain table	Enter `MHz, operating gain dB` rows	Uses the entered response within its supported domain
Additional selected-output loss	Enter only loss after the gain's measured output plane	Subtracts that extra loss; leave zero when the splitter/pad is already included in the measured net gain
Desired low/high output	Enter the station's required targets, or review defaults from corrected launch	Determines gain needed after the preceding cable/passives; a target is not measured output
Desired gain reserve	Enter the requested reserve in dB	Separates meeting a target from retaining adequate margin

Function	Inputs and use	Output and limit
Optional installed output limits	Enter verified limits for model, channel loading, tilt and output path	Checks requested/predicted output against those entries; blank means capability unchecked
Gain budget	Review required gain, entered gain, low/high margins and reserve status	Can show missing, invalid, shortage or marginal reserve. One good marker does not compensate for another bad marker
Manufacturer gain reference	Read the source, frequency/configuration and qualification	Screening reference only; a scalar rating does not become a flat gain curve or prove maximum available gain
Calculate gain from test-point readings	Enter input/output readings at the same frequency plus their separate remaining corrections	Corrected output minus corrected input; no gain is inferred from an output-only photo
Use as low / high gain point	Apply the helper result to the chosen gain marker	Updates the calculation input; does not align physical hardware

Complete's helper uses **7.1/16.1 dBmV +20 dB each** at 54 MHz: corrected 27.1/36.1 dBmV gives **9 dB gain**. A1's entered anchors are 9/18 dB at 54/860 MHz; A2's table is 54 MHz/8 dB, 550 MHz/12 dB, 860 MHz/15 dB. These are synthetic net gains, not GainMaker GaN control settings.

Installed equipment, manuals and control paths

Where: Route → Amp / node setup; also **Source setup & manual** from Setup. **Practice:** Complete; OEM AGC; DSIM; Manual/thermal; GS7K node; Six actives.

Function	Inputs and use	Output and limit
Equipment to set up	Select the launch node/active or a route active	Opens that exact station's saved draft and applicable reference context
Identity and manual review	Match model/module, split, port, firmware, manual revision and measurement plane	Displays source-qualified steps and references; a product brief is not a commissioning procedure
Setup frequencies & job levels	Read selected-profile base anchors, actual-carrier draft and measurement-plane reminders	Keeps design targets, estimated carrier targets, approved pilot and procedure backoff frequencies separate
Installed gain control	Select **OEM GainMaker AGC**, **ACI DSIM**, or **Manual / thermal control** where applicable	Selects the matching guide; changing hardware/control/pilot context can invalidate prior reviewed steps
Documented setup method	Choose the exact station's applicable manual-backoff or thermal branch	Supplies source-specific guidance. Never combine branches/revisions because a setting looks familiar

Function	Inputs and use	Output and limit
Reviewed steps and station notes	Record what was reviewed and unresolved	Persists with that station; not proof of work, pilot lock or RF acceptance
External source links	Open the original or explicitly labelled historical reference	Needs connectivity; verify exact installed applicability and document revision

Conventional GS7K: the node guide distinguishes receiver/FCM/RCM/routing, selected-port pads/EQs and the actual reverse transmitter/EDR/RPD. Conventional LCM Auto Set-Up discovers modules; it does not balance RF. Intelligent, SHO/Fiber Deep and retrofit-RPD references require their own hardware/procedure and are not interchangeable conventional-node profiles.

Complete's GainMaker GaN HGD: exact installed GaN commissioning/control instructions remain unresolved. The app keeps this gap visible. Conventional GaAs OEM pad/backoff formulas, ACI DSIM instructions and AOI alignment sequences do not establish GaN settings.

Focused OEM AGC: match the detector, approved pilot and manual. The displayed manual-backoff and thermal preparation branches are alternatives. Conventional GaAs system-amplifier and LE references use different pad relationships and different backoff tables; the station/manual must match. No universal OEM QAM detector offset or settling timer is supplied.

Focused manual / thermal: identify the installed control path and follow its own applicable method. Do not infer OEM AGC pilot-pad values or DSIM codes from that selection.

ACI DSIM functions:

Function	Inputs and use	Result and limit
Module / firmware / approved pilot	Identify DSIM variant, controller/module revision, pilot MHz/type and approved operating context	Keeps the setup bound to the installed equipment; DSIM-SG and later DSIM-A variants are not assumed equivalent
Physical-controller/app guidance	Read the exact cited interface/procedure and temperature preparation	Source-specific review only; physical RF adjustments and raw knob counts are not generic app dB settings
Historical pilot lookup	Choose channel or exact frequency and Analog/Digital type	Returns the literal ACI Rev A code when available. Channel 110 digital is 711 MHz, `@@?<A`
Copy code / manual fallback	Copy from a supported user action, or select the code for manual copying	No device is programmed; copying does not verify compatibility, a supervisor key or pilot lock
Missing/anomalous lookup rows	Read the unavailable/anomaly message	No codes are invented for 95–99, 303 MHz or OFDM PLC; channel 100 copying is blocked as a source anomaly

Function	Inputs and use	Result and limit
Key/review status	Record the review flag supported by the form	Do not enter or share an actual supervisor key; a flag is not credential verification

The historical Rev A table contains 86 literal rows: channels 52–94 and 100–142. DSIM-SG Rev E's pad instructions differ from Rev H; the app preserves that conflict rather than mixing settings. Match the full installed procedure before physical use.

AOI in the separate six-active example: HGD/HGBT guidance covers saved downstream profile → downstream Auto → optional downstream Manual → applicable upstream sequence → ALSC after both paths. The cited historical SA procedure is not a substitute for missing exact Quantum18 LE/BA I&O. Its instrument instructions remain separate from the local XPERTrak return workflow.

Equipment catalog

Where: Route → Catalog. **Practice:** Complete → Equipment references.

Choose category/family and search the exact model. The list includes RF equipment, cables, node profiles and power models. Search can widen beyond a family when that filter has no matching entry; read the displayed count/filter message and narrow a long list.

Expand **View published loss data**, **View operating gain & conditions**, **View bands & test points**, or **View power data & conditions**. Read units, frequency/voltage domains, source, configuration, nominal/maximum basis and source qualifications before using a value. Node and power entries link to their working setup/load screens.

Custom entries require actual input tables. Published references do not automatically fill installed operating gain. Unavailable curves do not become zero or extrapolate. UBT remains unavailable for new selections; older saved UBT definitions remain readable without silently replacing them. A legacy power row with unsupported current data needs an applicable entered load.

TDR distance locator

Where: Test → TDR. **Practice:** Complete → TDR; Custom curves → reverse TDR; Six actives → reconnect after A6.

Function	Inputs and use	Result and limit
Route endpoint/path	Select the actual endpoint whose feed path is tested	Excludes unrelated sibling branches
Connection point	Select Launch or the correct cable side after a route point	Changes the origin; route coordinates remain measured from the source

Function	Inputs and use	Result and limit
Direction	Choose toward endpoint or toward source	Supports forward and reverse testing along the selected path
Displayed distance and lead	Enter the meter's one-way displayed distance and physical test lead	Corrects the path distance; do not enter a round-trip time as feet
VF correction	Enter meter-set and actual velocity factors, enabling correction when needed	Scales displayed distance by actual VF / meter VF, then accounts for the physical lead
Locate result	Review corrected path distance, route coordinate, span and offsets	Places an entered distance on entered geometry; does not classify or detect a physical fault
Active boundary / invalid input message	Read the first barrier or missing/invalid condition	Uses the actual parent-chain order, including distinct active points sharing one route coordinate. A TDR path cannot cross another amplifier; reconnect on the correct cable side

Complete: after GaN A2 at **1700 ft**, **295 − 25 = 270 ft** of cable path locates **1970 ft**, **160 ft** into Main Tap B's 310 ft span. Both VFs are 0.87. Launch-origin testing beyond the active is blocked.

Custom curves: reverse from the 200 ft endpoint; **100 × 0.88/0.80 − 25 = 85 ft** toward the source places the hit at **115 ft**, 15 ft into the second span. These VFs and distances are teaching inputs.

Forward and return sweep setup

Where: Test → Meter setup. **Practice:** Complete for 42/54; Six actives for 204/258; focused equipment exercises for their matching procedures.

Function	Inputs and use	Result and limit
Split/workflow selection	Select 42/54 or 204/258 for the actual fleet/configuration	Retains separate setup drafts; changing a tab does not reconfigure hardware
Plant, equipment and reference	Record station/leg, equipment, instrument/firmware/options, service/sweep plan, reference, available band and measurement planes	Makes the comparison reproducible; instrument capability and compatible references still require verification
Guided setup	Use Previous/Next step and reviewed checkboxes; record actual settings/findings	Saved guidance/progress, not remote control or acceptance
42 MHz return sweepless	Use the approved DOCSIS-derived referenced response, actual service and matching saved reference	Comparison is in dB; no absolute physical receiver level or generator setting is inferred
Operator return TX reference	Record 38 dBmV separately, plus actual TX frequencies/service plan and transmitter/plane	A local level reference supplies no missing channel, return-injection or receiver target

Function	Inputs and use	Result and limit
Compare return markers	Enter actual low/high frequencies, readings, targets and tolerances in matching units	Reports differences and tilt against entered references; check the rest of the trace separately
204 MHz comparison basis	Choose referenced response (dB) or a verified absolute receive plane (dBmV)	Relative and absolute values are not interchangeable
Return sweep injection budget	For the 204 workflow, enter desired network-plane injection, total physical coupling/lead losses and applicable generator maximum	Estimates physical connector output and checks the entered limit; it is not automatically the ONX Injection field setting
Closeout notes and sources	Record findings, gaps and trace references; review original source links	No ONX/XPERTrak connection or equipment programming is performed
Save station snapshot	Save the station draft before editing another station	Adds a labelled snapshot to history, up to 100; preserves notes/readings, not acceptance
Saved station snapshots	Review history with station, split, time, leg/reference and training identity	Export includes snapshots and both current split drafts; it is not an external backup by itself
Export setup notes	Finish the text-file save/share action	Exports the entered setup, guidance and history; keep originals/full project separately

For physical-output budgeting, full physical losses remain losses even if instrument compensation is enabled. Do not enter the summed physical-output estimate as ONX network-plane Injection while also applying the same TPC. The Records injection helper instead uses **remaining uncompensated** losses for its manual-setting estimate. Neither establishes an approved meter mode or measured injection. Return Sweepless Sweep does not use this generator calculation.

Predicted sweep and output table

Where: Test → Predicted levels. **Practice:** Complete → Predicted sweep; Six actives → six restored marker outputs.

Function	Inputs and use	Result and limit
Trace selection	Select source and relevant route outputs	Shows the selected paths; compare through and tap ports separately
Signal level	Choose absolute dBmV mode	Predicted output versus frequency, dependent on valid source and component data
Normalized path response	Choose dB mode and reference MHz	Removes the launch shape and zeros the supported path response at that reference

Function	Inputs and use	Result and limit
Total path loss	Choose loss mode	Net path loss, including active effects where supported; not a measured sweep trace
View range and reference	Enter view start/end and supported normalization frequency	Changes the chart view/reference; does not change equipment capability or available spectrum
Pointer/touch readout and legend	Tap/move over the chart and read trace labels	Frequency/level readout for calculated data
Live/planning display	Review shading, solid/dashed presentation, gaps and chart notes	Above-live values are planning predictions. Gaps may be missing tables or launch support; never treat them as zero
Signal at each tap/output	Review the tabulated frequencies, ports and tilt/context	Tap results are at the F-port before the optional drop; fixed table markers can be unavailable on this route

Launch slope extension, when enabled, remains an estimate. Cable/passive coverage is not extended. Build 2 defers expensive chart redraw while this screen is hidden and draws current data when it is shown or printed. Frequency results are reused only within one synchronous refresh, not across later edits or imports. The model does not reproduce narrowband ripple, ingress, micro-reflections or a measured fault signature. A 993 MHz result cannot establish 1.2/1.8 GHz plant performance.

Single-path frequency checker

Where: Test → Frequency. **Practice:** Complete → Frequency; Custom curves → exact overrides.

Function	Inputs and use	Result and limit
Cable/passive combination	Select cable, feet, passive, actual path and quantity	A separate combination; does not add route points or alter RF topology
Exact frequency input	Enter a supported frequency from 5–1800 MHz	Returns the calculation and band/coverage flags at that point
Exact cable/passive overrides	Supply known dB/100 ft and loss-per-device at that frequency	Uses entered values already at stated conditions; clearing them falls back only to available reference data
Optional launch at that frequency	Enter a level at the matching plane	Computes receive budget after total loss; blank launch leaves receive level unavailable
Equipment/spectrum/passband limits	Enter installed, planned/live upper limits and split	Shows conflicting/out-of-band conditions; broad software range is not equipment support
Frequency table	Inspect low/high and other listed markers	Shows cable rate/loss, passive each/total, total loss, launch, receive and basis/status

Function	Inputs and use	Result and limit
Complete route / span summary	Select an existing route output	Summarizes that whole feed path and distance; different from the single combination above
Export frequency results	Export CSV and finish saving	Includes inputs, units, flags and unavailable values

Complete at **860 MHz**: **200 ft ×2.2 dB/100 ft + 1.4 dB =5.8 dB** total; **43 − 5.8 = 37.2 dBmV** receive. Custom curves at 261 MHz: 100 ft ×1 dB/100 ft +1 dB =2 dB, leaving 32 dBmV from 34 dBmV. These are exact-frequency teaching inputs.

Meter readings and prediction error

Where: Test → Readings. **Practice:** Complete → Meter readings; Six actives → A6 comparison.

Choose a route point and actual output, then enter frequency, reading, test-point status and remaining coupling. The comparison shows raw reading, corrected reading, prediction and **measurement minus prediction**. A negative error means the entered measurement is below the model.

Use zero/off for remaining correction when the meter already reports mainline-equivalent level. A reading at another frequency is not a valid comparison with the chosen marker. Missing predictions/readings remain unavailable. This task does not automatically fill the official forward/return records or approve their criteria.

Complete includes intentional mismatches. For example, A1's entered **16.1 + 20 = 36.1 dBmV** at 54 MHz is compared with about **36.94 dBmV** predicted. Tap C's direct **7.50 dBmV** compares with about **14.01 dBmV**. Investigate inputs and measurement planes; synthetic disagreement is not evidence of a physical defect.

Field references and target conversions

Where: Test → References. **Practice:** Complete → References; DSIM/OEM/GS7K → available carriers.

Function	Inputs and use	Result and limit
Your local non-high-split targets	Apply the labelled 31 @ 54 / 43 @ 860 · TX 38 preset to a compatible selected profile after confirmation	Updates its stated design/reference fields; does not write measured launch, approve criteria or supply missing measurements
Preset compatibility checks	Match hardware and forward passband containing both base anchors	Blocks high-split/incompatible use; a low return ceiling alone does not prove 54 MHz is usable
Setup targets at available carriers	Enter actual low/high channel centers and choose square-root frequency or linear MHz	Estimates targets from the original 54/860 anchors; range must satisfy 54≤low<high≤860 and known live/hardware limits
Fill 57/753 or 63/550 example	Fill only the converter draft	Does not replace base design anchors, measured launch or approved pilot

Function	Inputs and use	Result and limit
Common launch anchor pairs	Review/apply the labelled frequency preset	Changes its stated fields only; an example is not a live carrier survey
Editable node/launch target ranges	Use the selected profile or labelled workbook/sample draft; edit min/max/tolerance	Creates draft targets. **Apply ranges to selected profile** changes that selected profile, keeping approval separate
Published tap faces quick look	Select the route tap and frequency; open its face picker	Compares compatible published parts; choosing a face changes that route part
Cable loss quick look	Choose cable and frequency	Reference loss with source/domain qualifications; not an entered span measurement

Square-root examples from 31/43 dBmV at 54/860 MHz: **57/753 → 31.11/41.97 dBmV**; **63/550 → 31.32/39.79 dBmV**. The corresponding linear-MHz estimates are **31.04/41.41** and **31.13/38.38**. Curve choice requires the applicable local procedure; these are not a measured installed equalizer curve.

The available-carrier draft survives project backup and is reported separately in equipment setup/run sheet. It does not choose an AGC pilot. If actual live spectrum stops below the practice pilot, identify a compatible approved live pilot; the app does not invent one. The six-active Vecima/AOI profile keeps its separate in-band 261/993 MHz setup.

Field calculators

Where: Test → Calculators. **Added in:** 2.0 Alpha build 3 (in development). **Practice:** Custom curves → Quick calculations with source tags.

Each calculator shows its formula's source tag. Tap the tag for the source, revision, page or section, units, conditions and the date it was checked. A message instead of a number means an input is missing or out of range; it is never a zero.

Calculator	Enter	Result and limit	Source tag
dB converter	A level, its unit (dBmV, dBm, dBµV, mV or mW) and 75 Ω or 50 Ω	All five units. dBm = dBmV − 48.75 dB at 75 Ω or − 46.99 dB at 50 Ω; dBµV = dBmV + 60. RMS voltage across a matched load	Spec · decibel and SI unit definitions, re-derived
Ripple to distance	Spacing between ripple peaks (MHz) and the cable's VP	Echo path length between two mismatches: 491.79 × VP ÷ spacing (the published 492 is rounded). One mismatch may be the device you test at	Expert · Hranac, *Communications Technology*, Aug. 2005; SCTE 280 2022 section 9.7

Calculator	Enter	Result and limit	Source tag
Pre-EQ tap to distance	Echo tap, main tap (8 is typical; the CMTS sets it), symbol rate 5.12, 2.56 or 1.28 Msym/s, and VP	Spacing between two mismatches (not the distance from the modem or node), the ±½-tap range and feet per tap. Taps after the main tap only. DOCSIS 2.0/3.0 single-carrier upstream; not DOCSIS 3.1 OFDMA	Expert · CableLabs PNM Best Practices V03 sections 5.5, 6.6.2 and 6.8.1.1, on PHYv3.0 section 6.2.16 tap spacing
OFDM power	Measured power, the bandwidth it was measured in (6, 24 or 1.6 MHz) and the occupied bandwidth	Total power and power per 6, 24 and 1.6 MHz. Assumes flat power; a tilted output needs the total composite power method in SCTE 270 section 30.14	Spec · CableLabs PHYv3.1-I20 section 7.5.9.1; SCTE 258 2026
Pad / EQ balance	Amplifier model, low and high pilot (MHz), your readings and the station's targets at both, and the installed input pad and EQ	Level error at each pilot and the tilt error, with the direction to move the input EQ and pad. For GainMaker 1 GHz models it picks the input EQ for tilt, then the input pad for level at the high pilot, from Cisco's published values, and predicts the levels after the change. Quantum18 HGD and HGBT show that they align electronically in AOI's QuantumLink Local app; the LE shows that no public procedure or pad/EQ list was found; the BA gives pad direction in its 2 dB steps	Expert · Hranac, Broadband Library, Feb. 2022; GainMaker guide OL-31156-01 procedure order
Echo delay	Cable length between two mismatches and VP	Round-trip delay in ns and µs, and in 5.12 Msym/s symbol periods	Expert · Hranac, *Communications Technology*, Nov. 2009; speed of light from SCTE 293-2 2024 section 4.1

Checked worked numbers (the balance example above was also checked against a LibreOffice-recalculated spreadsheet): ripple 6 MHz apart at VP 0.87 → **71.3 ft**; tap 9 with main tap 8 at 5.12 Msym/s and VP 0.87 → mismatches **83.6 ft** apart (41.8–125.3 ft), or **167.1 ft** at 2.56 Msym/s; 42 dBmV per 6 MHz over 192 MHz → **57.05 dBmV** total; 100 ft at VP 0.87 → **233.7 ns**.

Pad / EQ balance: the readings and targets must be at the same two pilots and the same measurement plane. Put the station in the manual or backoff state its procedure requires before changing parts; an active AGC or ALSC loop can hide the change. GainMaker values come from Cisco OL-31156-01 (system amplifiers) and OL-31155-01 (line extenders) with S-A data sheet 716336 Rev D: input pads 0–20.5 dB in 0.5 dB steps and 1 GHz forward cable EQs 1.5–30 dB in 1.5 dB steps, with each EQ's published loss at 52–1000 MHz (the top column is headed 1002 MHz in data sheet 716336, so a 1002 MHz pilot is accepted). Loss between published frequencies is interpolated linearly in MHz, never across the chart's 86–550 MHz gap, so a pilot in that gap gets no pick. Pilots must be inside the model's forward band. The advice says whether to keep or change each part, gives the high-pilot level predicted after the EQ change, and says whether the result lands inside the procedure's ±0.5 dB window for tilt and level. The 0 dB EQ jumper has no published loss chart, inverse EQs have conflicting

charts, and 870 MHz EQs are for 870 MHz designs, so none of these are offered. Worked example: GainMaker HGD at 54 / 860 MHz, readings 36.4 / 49.6 dBmV, targets 39 / 49 dBmV, installed pad 6 dB and EQ 9 dB → EQ **4.5 dB** and pad **7 dB**, predicted **39.0 / 48.9 dBmV**. Balancing on two pilots does not replace a sweep.

VP: pick a catalog cable that lists a nominal VP (MC2 .440, MC2 1.000, Series 6 and Series 11 drop), or type the VP from the cable's datasheet as a fraction from 0.50 to 1.00. Most hardline catalog entries do not list a VP yet. For DOCSIS 3.1 downstream, occupied bandwidth is a whole number of 6 MHz channels, at least 24 MHz, and includes the edge taper.

The calculators do not save their inputs, change the job or read a modem, CMTS or meter. Results are calculations, not measurements.

Limits and sources

Where: Test → Limits, and the tag beside each formula. **Added in:** 2.0 Alpha build 3 (in development). **Practice:** Custom curves → Check a limit and its source.

The limits, targets and formulas behind the field calculators and this job's criteria, with where each came from. Each entry carries one of four labels:

Label	Meaning
Spec	A published rule, standard or definition: FCC rule, CableLabs specification, SCTE standard or SI definition
Expert	Published expert guidance; good practice, not a standard
Company	A local target entered by you or your company. It starts unknown until someone enters it
Unknown	The source is not verified; treat it as unconfirmed guidance

Tap a tag to open its source sheet: the value or formula, units, when it applies, which plant, the source, revision, page or section, the date it was checked and how it was checked, and what uses it. Filter the list by label with the buttons at the top.

What the list holds:

- The calculator formulas and the speed of light they use.
- Your built-in local preset: **31 dBmV at 54 MHz**, **43 dBmV at 860 MHz** and the separate **38 dBmV return TX reference** (42/54 plant).
- This job's project-profile criteria (design launch, level and tilt tolerances, ripple, MER, receiver and injection targets) and the meter-setup TX reference of the 42/54 workflow. The 204/258 workflow has no TX reference. Blank entries show **Unknown — not entered**; an entered 0 stays 0. New jobs start blank, except the 750 MHz example profile's labeled 32 / 43 dBmV launch levels. A company sheet links to the screen where you edit the value.

- The Night-log **200 entries/job** application guard, labeled **Unknown** under **Night log application guards**. Its source is the existing `night-log.js` software bound, not an RF or company acceptance limit.
- Two built-in approximations labeled **Unknown** because their sources are not verified: the 0.1 % per °F cable-loss temperature correction, traced only to an indexed copy of CommScope's MX0398 pocket guide, and the 0.004 per °C default power-cable resistance coefficient, which has no recorded source.
- Three software caps of the Outage card and its shared checklist engine, labeled **Unknown** and listed under **Checklist cards application guards**: up to **40 items** you add, **160 characters** in one item and **120 characters** in a note box (some boxes are shorter). Each has a tag beside the **Add an item** and **Notes for this job** headings. They are storage and display bounds, not company or RF limits.

The list describes values; it does not approve criteria or change the job. Example targets on older screens, such as the workbook and sample targets in Test → References, are not listed yet, and older screens do not show tags; later build-list tools add them where their values appear.

DOCSIS health card

Where: Test → DOCSIS health. **Added in:** 2.0 Alpha build 3 (in development). **Practice:** Custom curves → Check modem levels against the spec.

Compare a reading you already have, from a meter or a modem's status page, with what the DOCSIS specifications require a cable modem to accept or transmit, and with your own window. Nothing is read from a modem, CMTS or meter, and nothing is saved in the job.

The spec range is a capability limit, not a target. It is the range of levels a modem must be able to handle. A reading inside it does not mean the link is healthy: error rate, MER and the other checks are separate. A message instead of a result means an input is missing or out of range; it is never a zero.

Card	Enter	Spec comparison	Source tag
Downstream SC-QAM channel	The level of one 6 MHz digital channel at the modem input (dBmV)	−15 to +15 dBmV per 6 MHz channel; a reading exactly on an edge counts as inside	Spec · CableLabs PHYv3.0 C01 section 6.3.2, Table 6–22 (PHYv3.1 section 6.4.2 applies it to 3.1 modems)
Downstream OFDM channel	The OFDM power (dBmV) and what it was measured over, which you must choose from what your meter or modem page says: each 6 MHz, 24 MHz, or another occupied bandwidth (a whole number of 6 MHz channels, 24 to 192 MHz)	−15 to +15 dBmV per 6 MHz and −9 to +21 dBmV per 24 MHz, as printed. For another width the 24 MHz row moves by 10·log10(occupied ÷ 24), which is derived, shown to 0.01 dB and rounded inward so the range on screen is the range used. The card calls it a derived range and tags it Expert, not Spec, because the specification prints only the 6 and 24 MHz rows	Spec · CableLabs PHYv3.1 I20 section 7.5.11, Table 45

Card	Enter	Spec comparison	Source tag
Total downstream input power	The power of all downstream signals together (dBmV) and whether the modem is DOCSIS 3.1 or 3.0 only, which you must choose because the limits differ	Strictly below 40 dBmV for a 3.1 modem (54 to 1794 MHz) or 33 dBmV for a 3.0-only modem (40 MHz and up). Exactly 40 is not below 40. The 3.1 specification also points a 3.1 modem to the 3.0 SC-QAM table (33 dBmV total) and does not say which governs for SC-QAM-only reception, so confirm a 33 to 40 dBmV total on a 3.1 modem	Spec · PHYv3.1 I20 Table 45; PHYv3.0 C01 Table 6–22
Upstream transmit headroom	How the modem is working (a DOCSIS 3.1 modem on a 3.1 CMTS, or anything else), the modem's total transmit power (dBmV) and its Pmax	For a 3.1 modem on a 3.1 CMTS: headroom = Pmax − total, with Pmax 65 dBmV unless the modem reports more; 65 is the lowest the specification allows. It is a minimum capability, not a limit on what the modem sends. For a 3.0 modem, or a 3.1 modem on a 3.0 CMTS, the 3.0 per-channel limits apply instead; the card has no figure for them, so you type in the modem's own maximum from its documentation and the result is not a spec comparison	Spec · PHYv3.1 I20 sections 7.4.12.2 and 7.4.12.3.1

Each result says in words how far inside or outside the range the reading is, so it does not rely on color. The printed 6 MHz and 24 MHz OFDM figures differ by 0.02 dB (10·log10 of 4 is 6.02 dB, not 6); the card uses each as printed.

Company window. Every card has a **Company window** box for the lowest and highest level your company wants. Leave a side blank for no limit on that side: blank is unknown, never zero. The OFDM window is in dBmV per 6 MHz, and the card restates your reading per 6 MHz to compare. Windows start unknown and are saved on this device only, in the browser's storage, so they are **not** part of a job or its full backup; **Clear company windows** removes them for both plants. If the browser will not store them, the card says so and keeps them until the page closes. Company windows are kept **separately for each plant**: the box follows the plant chosen in **Test → Meter setup** for the job, so a window typed for the 42/54 plant is never applied to the 204/258 plant, which stays unknown until you enter one (no plant chosen: the boxes stay locked until you choose). They are shared by every job on that plant, and they also appear in **Test → Limits** under their own heading, one entry for each plant. A box holding text that is not a number (a lone minus sign, for example) is refused with a reason and never clears or replaces a saved window; for the upstream card it is refused instead of being read as a blank Pmax.

Total upstream power, not one channel. Enter the modem's total transmit power. In DOCSIS 3.1 operation the specification reports per-channel power normalized to 1.6 MHz, so a 6.4 MHz channel's actual power is about 6 dB above the number reported; your modem page or meter may use a different convention, so check which one it uses before adding channels up.

Checked worked numbers: 16 dBmV on an SC-QAM channel is **1.0 dB above** the range, and 3.2 dBmV is inside, **11.8 dB** from the upper edge; −8 dBmV over 24 MHz is inside, **1.0 dB** from the lower edge; 40 dBmV total on a 3.1 modem is **at** the limit, not below it; 52 dBmV total transmit power on a 3.1 modem with the 65 dBmV default Pmax leaves **13.0 dB** of headroom.

What this card leaves out on purpose: MER and SNR, per-channel upstream power, the limit on level differences between channels, DOCSIS 4.0, and the EuroDOCSIS 8 MHz ranges, which differ from the North American ones used here.

Leakage

Where: Test → Leakage. **Added in:** 2.0 Alpha build 3 (in development). **Practice:** Custom curves → Check and log signal leakage.

Five tools for signal leakage on one screen. **This is your own notebook and calculator, not the company's official leak record, and none of its results is a filing.** Nothing is sent anywhere. The leak log, saved detector checks and company settings are kept in the browser on this device only, under their own storage key. They are **not** part of a job or its full backup, and only the log can be exported (CSV); the saved checks and company settings cannot.

Tool	Enter	What it does	Source tag
FCC leakage limit check	Test frequency (MHz), analog or digital, the field strength (µV/m) and the distance from the leak	Picks the limit for the band and compares your reading with it at the rule's distance. The limits as the rule prints them: analog 15 µV/m and digital 13.1 µV/m at 30 m at 54 MHz and below and over 216 MHz; analog 20 µV/m and digital 17.4 µV/m at 3 m over 54 MHz up to and including 216 MHz. A reading exactly at the limit is within it. A reading restated from another distance gets a neutral mark and the word approximate, not a green check. It also says when your distance is inside the near field at that frequency	Spec · 47 CFR 76.605(c), Table 1; a separate row records the § 76.614 expectation for the aeronautical bands
Distance to 3 m	Field strength and distance in metres or feet	Restates the reading at 3 m and at 30 m: E at 3 m ≈ E measured × distance ÷ 3 m. In the far field the field strength falls in proportion to distance. Close to the leak it does not, and how close depends on the frequency: a common rule of thumb puts the edge of the near field at the wavelength ÷ 2π, about 0.4 m at 121 MHz and 4.8 m at 10 MHz. The card warns inside 3 m; the limit check, which knows the frequency, warns inside that edge. A restated reading is approximate	Expert · far-field relation; SCTE 221 2020 (R2026) for the far-field definition (secondary: read by a review agent). SCTE 293-12 2025 was identified from its catalog summary only

Tool	Enter	What it does	Source tag
Cumulative leakage index (CLI)	The field strength of each leak at 3 m, separated by spaces, new lines or semicolons (a comma followed by a space also works), and the fraction of the plant examined (or strand miles tested and total, not both)	10·log10 of (1 ÷ fraction examined) × the sum of each leak's field strength squared, counting only leaks of 50 µV/m or more at 3 m, compared with the printed value of 64. If no leak counts the card says so instead of showing zero. The fraction runs from 0.001 to 1. Under 75% examined the result is marked as an estimate only, with no green check, because the rule asks for at least 75% of the strand. A comma with no space after it, as in 1,500 or 120,5, is refused: it could be a decimal comma or a thousands separator. Strengths must already be on the 108–137 MHz basis: the FCC's frequency correlation is not done here	Spec · 47 CFR 76.611(a)(1) and (b)(1)
Detector check	The check method (a directly connected source, or a calibrated leak with a half-wave dipole), the expected and read values (µV/m) and the date (the box starts as today; a blank date is refused, never replaced by today)	The difference in dB and per cent, and whether it is within your company tolerance, which starts unknown. A saved check keeps the tolerance that was in force when it was saved, so changing the tolerance later does not change an old record	Expert · SCTE 289 2024r1 (secondary: only the SCTE catalog summary was read)
Leak log	Date found, where (pole, tap or cross street), probable cause, date repaired (blank while open), and optionally field strength, distance, frequency and notes	A list of leaks with open ones marked, the 3 m equivalent and whether it counts in the CLI, a **Mark repaired** action (the date box starts as today and can be changed; if you clear it, Mark repaired is refused and the leak stays open), **Reopen** to undo a mistaken Mark repaired, **Delete** and an **Export log (CSV)**. A leak counts in the CLI only if it is at least 50 µV/m at 3 m with the test frequency in the 108–137 MHz band; otherwise the row says what is missing. A leak logged with a field strength but no distance says it was not restated, compared with your repair threshold or checked for the CLI. The fields follow 47 CFR 76.1706 (date and location found, date repaired, probable cause; operators keep the log for two years, and for five years in some systems under 1,000 subscribers, § 76.601(d))	Spec · 47 CFR 76.1706

Company settings (a box at the bottom): your repair threshold in µV/m at 3 m, which flags open leaks in the log at or above it, and your detector tolerance in ± dB. Both start unknown; blank is never zero, and editing one never changes the other. They are one set for the whole device and appear in **Test → Limits** under their own heading. A box holding text that is not a number (a lone minus sign, for example) is refused with a reason and never clears a saved setting; the same goes for the number boxes of the log and the other cards, where a malformed entry is never read as "not given". The FCC limits are limits on the field a system may produce. For aeronautical-band leaks, § 76.614 separately expects an operator carrying signals there to note and repair, within a reasonable period, any leak of 20 µV/m or more at 3 m. The repair threshold is your company's own and starts unknown.

Keep the log free of customer information. Describe a place by pole, tap or cross street. Do not enter customer names, addresses or account numbers: the place and notes text go into the exported CSV. **Clear everything on this device** asks first and clears the log, the saved detector checks and the company settings; only the log can be exported before that.

If this browser will not save. A change this browser would not store stays on the screen until the page closes and is written with the next change that does save. If the saved notebook cannot be read, a copy of it is kept on this device and the screen says so before the next change replaces it. If a newer version of the app saved it, this version will not overwrite it.

Checked worked numbers: 25 µV/m analog at 3 m and 121.2625 MHz is **5 µV/m over** the 20 µV/m limit; leaks of 100, 60 and 50 µV/m with 0.8 of the plant examined give a sum of squares of **16,100**, an index of 16,100 ÷ 0.8 = 20,125, which is **43.04 dB**, 21.0 dB at or below 64; leaks of 120, 85, 60, 250 and 50 µV/m at 0.8 give **50.52 dB**; 40 µV/m at 10 m is **133.3 µV/m** at 3 m.

What this screen leaves out on purpose: the airspace measurement alternative, the correction for a test frequency outside 108–137 MHz, tracking of the quarterly aeronautical-band monitoring, the half-wave dipole voltage to field-strength conversion, per-job leak logs and live detector or GPS data.

Impairments

Where: Test → Impairments. **Added in:** 2.0 Alpha build 3 (in development). **Practice:** Custom curves → Find an impairment from what the meter shows.

Pick what the meter or spectrum display shows, from ten pictures, and read the matching card: **what it looks like**, **what usually causes it**, **what it is easily mistaken for** and **five steps: analyze, localize, prioritize, repair, validate**. It is your own aid and **not your company's procedure**, nothing is read from a meter or modem, and nothing is saved in the job or on the device. What a card says about the look and the causes follows SCTE 280 2022, paraphrased and cited by section; the five-step order is the workflow of SCTE's April 2026 PNM course series; the steps under each are general practice, labeled Expert.

You see	Card	Source
A regular ripple repeating across the spectrum	Standing wave	SCTE 280 §9.7 to §9.7.3
One notch spanning a few channels	Suckout	SCTE 280 §9.3
One hump that rises and falls, sometimes with a dip	Resonant peaking	SCTE 280 §9.5
Levels falling away gradually toward the top of the band	Roll-off	SCTE 280 §9.6
A steady slope across the band	Tilt	SCTE 280 §9.8 heading only; SCTE PNM series (body not read, so the card says so)

You see	Card	Source
Every frequency lower by about the same amount	Excessive flat loss	SCTE 280 §9.9 heading only; SCTE PNM series (body not read, so the card says so)
Narrow spikes at fixed frequencies, such as FM or VHF	Ingress	SCTE 280 §9.4.2, §9.4.4; SCTE 294 2024r1
Interference high in the downstream, above about 600 MHz	Cellular (LTE) ingress	SCTE 280 §9.4.3
A raised upstream floor or haystacks that go with the downstream signals	Common path distortion	SCTE 280 §9.4.7; SCTE 294 2024r1
Short wide-band bursts in the upstream	Impulse and burst noise, such as power-line gap noise	SCTE 280 §9.4.6; SCTE 294 2024r1

The standing-wave card finds two mismatches on your saved route. Enter the ripple spacing between neighbouring peaks (MHz) and the cable's velocity of propagation. The distance between the two mismatches is 492 × VP ÷ spacing, in feet; the card shows it with the same calculator as Test → Calculators and lists the pairs of points on the saved route whose cable distance is within a search width of it (10 percent unless you change it for that search). A pair is the launch point or a point's far end of a span and a later point on the same path, so a candidate can span several cables; parallel branches are never added. Each candidate is a place to look at both ends, not a finding: a mismatch can be a connector, splitter, tap, amplifier input or the cable itself, and the real pair may lie outside the saved route. The search width is registered in Test → Limits as an Expert value; no source sets it.

Checked worked numbers: a 1.7 MHz ripple on 0.87 VP cable is **251.7 ft** and, on the sample route, points to the 250 ft span from Tap 1 to Tap 2; a 4.3 MHz ripple on 0.87 VP cable is **99.5 ft** and, on the custom-curves practice route, points to its two 100 ft spans.

The list follows the working job. It is recomputed whenever the route or job changes while this screen is open, including Files → Open project and switching to another job; opening a project that has no route clears the old places and says the route has no point with a valid distance. Each number box has its own label, and the two Expert tags beside the heading and the search width are separate 44 px controls.

What this screen leaves out on purpose: reading peaks off a sweep, any live capture, amplitude or depth limits (those are your company's), PNM data from modems, impairments of the node or headend, and the cards for amplifier distortion noise, HPNA, OBI, filters and missing channels, pullout and adjacent-channel alignment (named in the SCTE series but not built here).

Fiber

Where: Test → Fiber. **Added in:** 2.0 Alpha build 3 (in development). **Practice:** Custom curves → Find a fiber and estimate a link loss.

Two small fiber tools. Offline; nothing is read from a meter and nothing is saved. A blank entry is never read as zero.

Fiber color finder. Counting from 1 across the cable with 12 fibers in each tube, enter a fiber number (1 to 288), or a tube (1 to 24) and a position inside it (1 to 12). It shows the fiber's color and position and the tube's color, using the TIA-598 order for both fibers and tubes: blue, orange, green, brown, slate, white, red, black, yellow, violet, rose, aqua. Past tube 12 the colors repeat with a stripe or dashes and makers differ (tube 20 is the black position, and makers mark the repeat differently: a white tracer in one maker's note, a yellow stripe in another's), so the cable print legend and the construction sheet decide. Cables with 24 fibers in a tube, ribbons and other color schemes are not covered. The source is a secondary summary of ANSI/TIA-598: the standard itself was not read.

Link loss budget. Estimated loss = length × attenuation + mated connector pairs × loss per pair + splices × loss per splice + other loss (a splitter or WDM, from its data sheet). A count of 0 needs no loss; a blank count, or a blank loss for a counted item, gives a reason and no number. Add the safety margin to get the budget the link must fit in. With the transmitter output power and the receiver sensitivity (the weakest signal it accepts, both in dBm) it also shows the optics budget (transmit power minus sensitivity) and how many dB are left, or the shortfall. Without them it says it is the loss to expect, not a pass or fail. It does not include measurement uncertainty and does not judge a link against your company's pass limit, which the app does not know.

The standard's allowances are shown, never filled in silently. A table lists TIA-568.3-D's worst-case design values: 0.75 dB per mated connector pair, 0.3 dB per splice, and for single-mode cable 0.5 dB/km outside plant or 1.0 dB/km inside plant at both 1310 and 1550 nm. Each has a source tag; they are labeled Spec and flagged secondary, because they were read in vendor and association summaries and not in the standard, and TIA-568.3-E (2022) replaced the -D edition and was not checked. Real glass and connectors measure better (about 0.35 dB/km at 1310 nm, 0.2 to 0.5 dB per pair), and your cable maker's datasheet is the better number. **Copy the outside-plant allowances into the fields** and **Copy the inside-plant allowances into the fields** put the standard's values in the boxes when you ask; nothing starts with a default.

An unfinished number is reported where you typed it. A number box that holds only a minus sign or an unfinished exponent is never read as blank and never replaced by another way of entering the same thing. The box turns red, a message directly under that box names it and says to finish the number or clear the box, the box points at the message for a screen reader, and the result below says it is not calculated.

Checked worked numbers: fiber 30 is white (position 6) in tube 3, which is green; fiber 240 is aqua in tube 20 (black, with the marking noted); a 12.4 km link at 0.35 dB/km with 4 connector pairs at 0.3 dB, 6 splices at 0.1 dB and 3.5 dB of other loss is **9.64 dB**, **12.64 dB** with a 3 dB margin, and with a 3 dBm transmitter and a −24 dBm receiver the optics budget is **27 dB** with **14.36 dB** left.

What this screen leaves out on purpose: OTDR events, multimode budgets by wavelength, a bandwidth or dispersion budget, PON split budgets, polarity and MPO fiber maps, fiber-count conventions with more than 12 per tube, and any company pass limit.

High split

Where: Test → High split. **Added in:** 2.0 Alpha build 3 (in development). **Practice:** Custom curves → Read the band plans and the Remote PHY bring-up steps.

A reading aid for the 204/258 MHz plant and for what is public about Remote PHY (R-PHY) bring-up. It takes no input, reads nothing from a node, meter or core, and stores nothing. **Every statement names its document, the issue it was read in and where in it.** It is not a procedure and does not replace your node maker's guide or your company's.

Band plans. A table of the DOCSIS 4.0 frequency-division-duplex band edges, in MHz: sub-split and extended sub-split (upstream 5 to 42, downstream starts 50 to 54 and ends 1584 to 1590), mid-split (5 to 85, 108 to 1644), high-split (5 to 204, 258 to 1794) and the ultra-high splits (upstream to 300, 396, 492 or 684, downstream from 300 to 372, 396 to 492, 492 to 606 or 684 to 834, all ending at 1794). The source is CableLabs CM-SP-PHYv4.0 issue I08 (September 28, 2026), Table 29 and Table 31. The 42/54 and 204/258 plants are different plans and stay separate. A second table gives the gap between the two directions (8 to 12 MHz, 23 MHz and 54 MHz): it is our subtraction of the printed edges, labeled derived, and the specification calls such spans transition bands, not guard bands. A short list gives what a DOCSIS 3.1 modem must support (CM-SP-PHYv3.1 issue I20, April 19, 2023): upstream edges from 42, 65, 85, 117 and 204 MHz, a 258 MHz downstream lower edge, demodulation to 1218 MHz, and 1794 MHz only as an option. No document read gives 102 MHz as a mid-split downstream edge; the specification says 108. SCTE 300 2024 (Best Practices for Migrating to DOCSIS 4.0) covers readiness and says it does not define band edges; SCTE 294 2024r1 was found only as a catalog entry. Passing a sweep to 993 MHz does not certify a plant for 1.8 GHz. The table assumes a contiguous occupied downstream spectrum, as its caption in the specification says; that is not a requirement, a device can leave a hole, and the downstream lower edge of the newer splits can sit anywhere from the upstream upper edge to the edge printed.

The aeronautical band and leakage. A table says which direction 108 to 137 MHz is in for each plan: downstream on the 42/54 plant and on a mid-split, **upstream on the 204/258 plant**, where it carries subscribers' modem signals (strongest at the home, weakest at the node), so a downstream test tone no longer stands for that band. The approach two SCTE papers (2020 and 2021) describe is a test burst the modems send on request (an OFDMA upstream data profile, OUDP) in a region the operator chooses just above the band, with the field detector set to match what operations configured on the CCAP; a maker's page and CableLabs' OSSI section 9.4 are named as pointers. The screen gives no detector setting, test frequency or pilot code, and it says no document read tells a technician whether to inject a downstream tag into that band on a 204 MHz node: ask your leakage group. The rules are in plain words with their sections: 47 CFR 76.610 (which operators the rules reach), 76.611 (a cumulative leakage index of 64 or less, or an airspace measurement not above 10 µV/m at 450 m), 76.614 (monitoring every three months, finding a 20 µV/m leak at 3 m, repairing within a reasonable time) and 76.616 (no strong carrier within 100 kHz of 121.5 MHz and 50 kHz of 156.8 and 243.0 MHz). Official leakage results belong to your engineering group.

Remote PHY bring-up. Eight steps the CableLabs specifications give for an RPD after power-up, listed in the order each is first needed (they are not one strict sequence: timing is synchronized in parallel with the principal and auxiliary cores' configuration, and has to be started before the core moves the RPD to operational), each with who does it and its document and issue: local initialization, network authentication,

address assignment, time of day, the control connection to the principal core, the data tunnels, timing, and the move to operational. It is automatic and most of it is done with the network and the core, so none of it is a meter check at the node. **Vecima nodes:** Vecima keeps its installation and configuration guides in its Customer Portal (a login), and only product briefs are public, so the screen does not describe a Vecima procedure and does not borrow another maker's steps or lights. Cisco, Teleste and Harmonic public guides are listed so you know they exist; their lights and steps are for their own hardware. When you have the Vecima guide your company holds, write its title and revision on your run sheet.

What this screen leaves out on purpose: a Vecima commissioning checklist (no public document), optical receive levels and timing-lock indicators (no public figure or display found), leakage detector and OUDP settings, pilot patterns, timers and event numbers, and a procedure for converting a node to high split.

Why cards

Where: Test → Why cards. **Added in:** 2.0 Alpha build 3 (in development). **Practice:** Custom curves → Read a why card and follow its link.

Fourteen short cards on the basics behind the numbers in this app, one for each "What is ...?" monograph in SCTE's 293 series. **Each card is our own short summary, not a copy**, and names the monograph to read in full: SCTE 293-1 2024 (the decibel), 293-2 2024 (velocity of propagation), 293-3 2024 (return loss), 293-4 2024 (loss, or attenuation), 293-5 2024 (unity gain), 293-6 2024 (skin effect and skin depth), 293-7 2025 (long loop automatic level control), 293-8 2025 (S-parameters), 293-9 2025 (RF), 293-10 2025 (signal level), 293-11 2025 (noise figure), 293-12 2025 (signal leakage field strength), 293-13 2026 (carrier-to-noise ratio) and 293-14 2026 (impedance). SCTE makes its standards free to download. Tap a card's name to open it: it says what the topic is, why it matters in the field, and which screen of this app uses it. A **Route → Station**, **Test → Calculators**, **Test → TDR**, **Test → Leakage** or other button under *In the app* takes you there. Topics with no screen in the app (S-parameters, noise figure, carrier-to-noise ratio) say so.

The cards hold no figure that is not either a definition or a registered one (the 75 ohm nominal impedance is a labeled secondary value in **Test → Limits**). Nothing is typed, stored or read from a meter. The titles and years are the ones in the SCTE Standards Catalog; for 293-7, 293-13 and 293-14 only the SCTE catalog or Standards Updates description was read, so check the monograph itself.

Proof of performance

Where: Test → Proof of performance. **Added in:** 2.0 Alpha build 3 (in development). **Practice:** Custom curves → Read the FCC proof-of-performance limits.

A reference for the technical requirements of the FCC's cable rules (47 CFR Part 76) and how an operator's proof-of-performance tests are measured against them. **It takes no input, stores nothing, makes or records no test and is not a filing. Official results belong to your engineering group.** The text is from the rules as read on the eCFR on November 11, 2026; every row names its rule section, the 16 rows that state a limit or a method also have a source tag (the three rows about QAM channels, test conditions and the test report name their rule section but have no tag), and the limits are registered in **Test → Limits**.

Four cards. **How often, where and on what** (76.601): tests at least twice each calendar year at intervals of no more than seven months; six widely separated test points for a system of 1,000 to 12,500 subscribers and one more for each further 12,500 or fraction, with at least a third at the most distant terminals; five channels below 550 MHz of activated capacity and ten at 550 MHz or more (the visual level, its variation and the aural level on every analog channel); systems under 1,000 subscribers are outside the performance-test paragraphs but, if they use spectrum beyond over-the-air TV and FM, must still do the leakage tests and monitoring, with the log kept five years. **The analog-channel limits** (76.605(b)): visual level not less than 0 dBmV at the terminal and +3 dBmV at the end of a 100 ft drop; level variation of at most 8 dB over six months with the adjacent-carrier and any-other-channel limits; aural carrier 4.5 MHz ± 5 kHz above the visual and 10 to 17 dB below it; frequency response within ± 2 dB; carrier to noise not less than 43 dB; coherent disturbances not less than 51 dB (47 dB); terminal isolation not less than 18 dB; hum at most 3 percent; and, every third year at the headend, chroma delay, differential gain and differential phase. **Digital channels and leakage:** QAM video follows ANSI/SCTE 40 2016, which is not read or listed here, and leakage is measured by the half-wave-dipole method of 76.609(h) against the table in 76.605(c) (see **Test → Leakage**). **How it is measured and kept** (76.609 and 76.1704): the system-noise method over a 4 MHz band, the test conditions, what the report must include and the five-year record.

These are the analog (NTSC or similar) limits. Many systems have little or no analog left, and what applies to yours and your franchise is for engineering to say. SCTE 204 2025, *FCC Proof-of-Performance Checklist for Analog and Digital Signals*, is the industry checklist for this and for how to carry out the tests; it was found in the SCTE catalog and **not read**.

Study mode

Where: Test → Study. **Added in:** 2.0 Alpha build 3 (in development). **Practice:** Custom curves → Answer a practice question.

Practice questions made from this app's own calculators and reference screens, grouped by four SCTE Broadband Distribution Specialist domains: **architectures; distribution components; signal types; maintenance and troubleshooting**. **They are original, not SCTE or NCTI questions, and not an exam.** Every number comes from a seed on a fixed grid, and the answer comes from the same function the app uses (the dB converter, the OFDM power, ripple-to-distance, the leakage restating and cumulative leakage index, the fiber budget and color finder, the DOCSIS range check, the high-split table, the impairment cards and the Outage checklist), so the question and the answer cannot disagree with the screen it practices.

The 18 question generators cover the four domains. Choose a domain (or all four), read the question, type a number or pick an answer and tap **Check my answer**. A right answer says so; a wrong one shows the right answer, the working and a button to the screen that does the calculation. **Next question** moves on and **New numbers** asks the same kind of question with different values. A blank answer, a lone minus sign, text or an exponent is never taken as an answer: the screen asks for a plain number and does not score it. Numeric answers allow a small stated tolerance (for example 0.05 dB or 0.5 percent). A score is kept for this visit only: **nothing is saved**, in the job or anywhere else, and a score here is not a test result.

The Outage question is grouped under maintenance and troubleshooting and asks which section of this app's own card a prompt belongs to. The questions do not follow any exam blueprint.

Supply PM

Where: Power → Supply PM. **Added in:** 2.0 Alpha build 3 (in development). **Practice:** Custom curves → Record a supply check and a span drop.

A personal notebook of what you saw at one standby supply, and a span-by-span voltage drop. **It is not the company power record, it does not decide pass or fail, and it is not saved in the job or in a project backup.**

The drop is span current times the cable catalog's published loop resistance, with the same temperature factor Power → Load already uses. You type the coefficient. **0.004 per °C is not filled in**; it is the engineering approximation named on Power → Load, and you type it only when you mean to use it. A cable with no published loop resistance is not calculated. The voltage at the end of the span is compared with the published minimum input of the active you pick from the catalog. If that model has no published minimum, the screen says so. **No company minimum is filled in.** Adding a span sets the next span's start to the previous end, so a chain walks span by span. SCTE 238 and the 2025 and 2026 SCTE powering projects named in the build list were not read.

The notebook records the date, input and output volts, load, an inverter observation (ran, did not run, not tested, or not recorded), chemistry as a label only (VRLA, lithium, other, or not recorded), alarms as you saw them, and battery-string volts. Two dates of the same string show the change in volts. There is no battery window and no inverter limit. Tap Save this visit to keep it on this device. Tap Delete this supply to remove that notebook. The job is not changed.

Power-supply load

Where: Power → Load. **Practice:** Complete → Power; Six actives → explicit power topology.

Function	Inputs and use	Result and limit
Supply voltage/rating	Enter coax output V RMS and A RMS rating	Defines the supply-side calculation and loading/headroom; not utility-mains current
Add equipment / junction	Enter each powered device once; use a zero-load junction for branches	Builds the actual power circuit independently of RF route parents
Power parent	Choose supply or the actual upstream power point	Shared sections carry downstream branch currents; RF direction alone does not establish this
Cable and resistance	Enter feet, supported cable or custom loop resistance and extra/contact resistance	Calculates section voltage drop and resistive loss. Custom resistance is used as entered at actual temperature

Function	Inputs and use	Result and limit
Power-cable temperature/coefficient	Enter the power temperature and applicable resistance coefficient	Adjusts supported nominal loop resistance; separate from RF cable temperature
Manufacturer current curve	Select the exact supported configuration	Interpolates current within its documented voltage domain; no borrowed curve or out-of-domain extrapolation
Entered amps	Enter known/measured current	Uses that load directly; synthetic examples are not manufacturer specifications
Entered input watts / PF	Enter watts and valid power factor	Estimates current from equipment voltage and PF; unsuitable/inconsistent inputs prevent a trustworthy result
Entered input VA	Enter apparent power in VA	Estimates current as VA divided by equipment voltage; power factor is not applied a second time
Minimum equipment voltage	Enter the applicable installed limit if known	Supports voltage warnings; a curve's table boundary is not automatically an undervoltage cutoff
Supply summary	Read total output current, loading %, headroom, wire loss and minimum equipment voltage	Scalar RMS planning result, not a measured quasi-square-wave waveform
Per-section table	Read equipment V, device A, section A, section drop V and loss W	Distinguishes a device's current from all current carried through its incoming section
Remove / edit row	Change only the intended circuit point and inspect child feeds	Verify actual power passing, supply boundaries and connector ratings independently
Export power load	Save the power CSV	Includes circuit inputs, configuration/source and warnings; not a full-project backup

Complete uses a **90 V / 15 A** supply; synthetic entered **0.75 A GS7K**, **0.60/0.45 A GaN** loads and a **20 W / PF 0.9** branch. Its independent reference result is **2.047 A**, **13.65%** loading and **88.83 V** minimum at GaN A2. The GaN selection has no verified current curve; no AOI curve is substituted.

Power what-if

Where: Power → What-if. **Practice:** Complete → Power what-if.

Choose the proposed power parent, cable, added length, current and minimum voltage. Read the trial's displayed **Predicted end voltage**, **Predicted voltage headroom** and status before applying it. The preview recalculates the copied circuit internally; it does not display total trial supply current. This calculation uses a copied circuit until you choose **Keep in power circuit**.

Keep adds the proposed point to the working power plan. Open **Power → Load** to review the revised total supply current/loading and all section results, then export the revised job. Resetting the training exercise restores its baseline after preserving the edited copy; for real work, maintain a deliberately saved/exported baseline before keeping trials.

Complete's trial adds **300 ft P3 .625 and 0.75 A after GaN A2**. The independent reference answer is **2.798 A** supply current, **18.65%** loading and **87.37 V** at the new load. These are planning estimates for this topology, not a general cable/device allowance.

ONX files, original evidence and comparisons

Where: Job → ONX files. **Practice:** Complete → Synthetic meter files; backup step.

Function	Inputs and use	Result and limit
Open project backup	Select a full NightSweep project JSON	Opens an entire saved job. To add evidence to the current job, use the file-attachment action instead
Evidence classification	Choose Sweep, Ingress, OneCheck or supported attachment context; stage Before, After or Reference; asset/station and notes	Labels the original file and makes before/after comparisons interpretable
Add meter files or photos	Choose available files and wait for completion	Stores original bytes with the job; does not control a meter or read an image's numbers automatically
Supported CSV preview	Supply explicit frequency and level/response/channel columns	Decodes supported interchange data; proprietary ONX JSON/XML/ZIP/PDF and other native schemas remain originals unless supported
View decoded measurements	Expand the parsed trace/channel table	Inspect values, units and source context; previewing applies no silent TP correction
Sweep / ingress before-after	Match compatible asset, kind, units and frequencies/stages	Compares entered file data; does not invent values or operator acceptance thresholds
OneCheck/channel table	Review supported channel, frequency, level and MER columns	Preserves discrete channel data; SC-QAM and OFDM PLC/block quantities are not interchangeable
Original-only attachments	Retain unsupported formats or photos	Preserves evidence without pretending it has measurement decoding or OCR
Save original	Export the exact stored original	Finish the device save action and verify the destination file
Verify saved files	Check current-job originals for missing/tampered bytes	Verifies stored file integrity, not a separate external backup or physical measurement

Function	Inputs and use	Result and limit
Remove from job	Remove the intended attachment reference	Protect/export evidence before removing it; removal is not a backup
Clear unused stored files	Clean originals no longer referenced by working/named jobs	Validates references first and stops conservatively on corrupt named-job data; do not use storage clearing as fault recovery

Supported interchange examples:

```
frequency_mhz,level_dbmv
54,31
860,43
```

Use `response_db` for relative dB response; a channel table can use `channel,frequency_mhz,level_dbmv,mer_db`. Provide complete finite values, explicit units and valid unique frequency/channel rows. A filename in a record is only a reference; attach the bytes if the original must travel with the backup.

Evidence/project limit	Maximum
One original attachment	8 MiB
Total original evidence in one job	12 MiB
Attachments in one job	30
Text decoded for a preview	2 MiB
CSV data rows for decoding	10,000
Full project JSON import/export	20 MiB

An oversized/unsupported preview can remain an original attachment within the original-file limits. Every project-opening route applies the common **20 MiB UTF-8 text limit before parsing or changing the working job**. Oversized or non-text input is rejected. A missing or tampered original blocks a portable export rather than silently dropping evidence. Wait for imports/exports to finish before switching jobs.

Field checklists and unit conversion

Where: Job → Checklists. **Practice:** Complete → Field checklists.

Function	Inputs and use	Output and limit
Leakage / ingress quick reference	Review shielding, fittings, unused ports, drops/premises and retest guidance	Definitions and inspection prompts, not leakage thresholds or automatic fault classification
Leakage repair checklist	Mark fittings, shielding/housings, terminations, affected drop/premises and retest items actually reviewed/performed	Saved work checklist; not RF acceptance
Leakage/ingress notes	Enter location, findings, repair, retest and trace references	Included with the job and relevant reports
Preventive maintenance / EOL	Record supply inspection, bonding, EOL points, amplifier/unity-gain and patrol notes	Individual checkmarks plus findings; route predictions are not copied into measured notes
dBµV ↔ dBmV	Enter the voltage level and starting unit	dBmV = dBµV − 60; reverse adds 60. Same reference plane; no TP addback or µV/m field-strength conversion

Practice **72 dBµV =12 dBmV**. This is a voltage-reference conversion, not dBm power, leakage field strength or proof of a measured level.

Outage first response

Where: Job → Outage. **Added in:** 2.0 Alpha build 3 (in development). **Practice:** Custom curves → Walk through an outage first response.

Prompts for the first steps of an outage call. **It opens with "Follow your company's outage procedure first" and it is not your company's procedure.** The wording was written for this app from general field practice, with no NCTI or SCTE text copied. It has **not** been reviewed by a supervisor against the local outage procedure, and that review is needed before coworkers use it.

Section	What it prompts
Confirm the scope	What is down (a tap or leg, an amplifier or node, a wider area), nearby nodes, planned work, utility power or weather, and the dispatch notes, with no customer names or addresses copied in
Power	Utility AC at the power supply, its status on utility power or battery and alarm lights, and telling your supervisor or the ROC when it is on battery or off
Node and optics	The node's status lights and alarm, the optical input and return path indicator against the value your company expects, and what to do when there is no light
Actives, in order	Working outward from the node one active at a time, noting the last that works and the first that does not, then the passives and cable between them

Section	What it prompts
Tell the right people	The ROC or dispatch (what, where, when), your supervisor, and a damage, theft or vandalism report before you touch anything
If damage or theft is suspected	Safety first, leaving the scene as it is until your procedure says otherwise, and writing down what you found, where and when, leaving customers out
Record the times	When you arrived, found the cause, told the ROC and when service came back, what you found and did, and confirming restoration with the ROC

Your notes (below the list): a plant place such as node leg 3 or amp 2, four times on a 24-hour clock (arrived, found the cause, told the ROC or dispatch, service restored), what you found and did, who you told as a role, and three optional damage or theft notes (what was found, who was told as a role, and the time it was reported). **Enter plant places and roles only: no customer names, addresses, phone numbers or account numbers.** The card has no field for them and holds no photos or locations. SCTE's 2026 Critical Infrastructure Protection Subcommittee is still writing an evidence-collection standard for damage and theft; this card does not implement it and only reminds you to follow your company's procedure.

Change this list lets you reword, hide, add and restore items. **Start a new call** unticks everything and clears all the notes (it asks first); your changes to the list stay. **Save as text** prepares a short note of the ticks and your notes to check before you share it. **Clear everything on this device** asks first and cannot be undone.

Where it is kept: in this browser on this device only, under its own storage key, outside every job and backup. A change this browser would not store stays on the screen until the page closes and is written with the next change that saves; unreadable data is copied aside before it is replaced, and data saved by a newer version is never overwritten.

What this screen leaves out on purpose: numbers, limits and levels (those are your company's), outage procedures, ticket or customer lookups, photos and locations, and any live read of a node, modem or amplifier.

Project records and closeout

Where: Job → Acceptance. **Practice:** Complete → Records; DSIM → findings; focused equipment guides.

Identity, profiles and acceptance criteria

Record ticket/node/design revision, lead, meter/calibration, reference ID, reviewer, acceptance date, disposition and notes. Job ticket/node/lead are linked with the corresponding Details fields.

Select the actual profile and enter **installed equipment upper**, **planned upper**, **available spectrum**, return ceiling/forward floor, forward/return marker frequencies, design or receiver targets, level/tilt/ripple tolerances, minimum MER, network injection targets/tolerance, control/procedure and marker basis. Mark approval only when the operator's criteria are genuinely approved.

A profile's name/container is not proof of its entered operating limit. For Complete, the profile container is labelled 1002 while the actual entered hardware/planning upper is **1000 MHz** and live upper **860 MHz**. Forward comparisons check their hardware, planned/live limits and valid ordered markers. Return comparisons also reject overlapping return/forward passbands. Resolve the reported band issue before interpreting a within-limits result.

Forward measurements

Use **Add** for a separate station/port/stage record. Enter date, node/station, profile, Before/After stage, **Planning or Measured** basis, measurement plane, actual low/high markers, raw readings, remaining addbacks, targets, ripple, trace/reference, trace-saved status, reviewer and notes. Forward rows also record model, MER, input pad/EQ, interstage trim and control system.

Read corrected low/high levels, tilt, target errors, tilt error, coverage and status. Planning data or unapproved/incomplete criteria cannot establish acceptance. A few passing markers do not certify the full trace, all frequencies or physical alignment.

Return measurements and injection setting

Return rows are for an **injected return test with a verified absolute receiver plane**. Enter the common station/stage/plane data plus injected low/high network levels, existing pad, ingress check, EQ and injection source. Referenced DOCSIS return sweepless belongs in Sweep setup, in matching dB units; do not disguise it as an absolute receiver record.

The estimated pad follows the entered receiver error and existing pad. A negative estimate calls for gain/path review; it is not silently changed to zero. Keep the separate **38 dBmV return TX reference** distinct from the chosen receiver targets and network injection levels.

The Records **Return injection setting** helper uses desired network-plane injection plus **remaining uncompensated coupling and lead** for a manual-setting estimate. Practice **40 + 20 + 1 = 61 dBmV** as form arithmetic only. This does not establish an ONX limit, physical output with compensation enabled, or a setting for the 42 MHz Return Sweepless Sweep method. The 204 Sweep setup physical-output budget uses total physical losses and has a different purpose.

Repairs, DSIM and record management

Function	Inputs and use	Output and limit
Repairs / punch list	Item, station/port, category, finding, action, owner, due/repair dates, retest, status and reviewer	Closed without required repair/retest/reviewer information remains incomplete
GainMaker / ACI DSIM record	Exact station/module/revision, plant/route, modes, pilot/type/lock, fallback, temperature, S1/pad/backoff/TGC, final verification and technician confirmation	Checks completeness/consistency and actual calendar dates for the supported DSIM-SG review; impossible imported dates remain invalid without rewriting the saved value. This is not a universal GaN/AOI/DSIM-A setup procedure

Function	Inputs and use	Output and limit
Add / remove record	Add a row or remove the intended record using the controls shown	Enter and review the actual identity, stage and measurement plane for each row
Suggest trace filename	Use the button on a forward/return row to generate a suggested name	Fills the trace reference and sets trace-saved status to **No**. It does not create, save or attach a meter file
Export section CSV	Export forward, return, punch or DSIM rows with their computed/status fields	Section report only, not the full project or original evidence
Summary and disposition	Read total records, After rows within entered limits and open/incomplete repairs	Reviewer disposition remains an explicit decision; no software calculation certifies the plant

The synthetic DSIM form in Complete is a separate form exercise. Use the dedicated DSIM job for its matching setup path and preserve the exact revision conflict. A completed setup record remains separate from forward/return acceptance.

Run sheet, exports and full backups

Where: Job → Run sheet; Files menu; relevant calculator/record task. **Practice:** Complete → Run sheet and exports / Backup.

Run sheet combines job/work-order context, selected frequency, source/model/split/limits, RF predictions, entered meter comparisons, as-built notes, field work, equipment/sweep setup and available-carrier estimates. Review its synthetic/sample warnings and missing values before sharing.

Output/function	Where and use	What it preserves
Print run sheet	Run sheet → Print	Printable field summary/PDF via device print service; actual pagination is device-dependent
Share	Run sheet → Share	Text through Android/browser share where supported, or the available prepared-file/download fallback; choose a recipient/app yourself
Export sheet CSV	Run sheet	Reviewable summary with context; not a full backup
Export RF levels	Files	Route predictions at listed frequencies; unavailable results blank
Export frequency results	Test → Frequency	Single-path inputs, frequencies, units, flags and results
Export power load	Power → Load	Power topology, loads, per-section results and qualifications
Record CSVs	Job → Acceptance	The selected section's entered rows and computed review fields
Export setup notes	Test → Meter setup	Setup drafts, guide/context and saved station snapshots
Save original	Job → ONX files	Exact original attachment bytes

Output/function	Where and use	What it preserves
Back up project	Files or Home	Full supported route, power, job, records, field/setup drafts, training state and original evidence
Open project / recover backup	Files, Home or ONX project shortcut	Validates and restores the entire supported project; verify route and originals after opening

Your file is ready means bytes are prepared. Choose **Save/share file** or **Download file** when offered, complete the phone's final save action, then confirm the result in Files/Downloads or the chosen destination. Cancellation is not a completed export. A PDF, screenshot, CSV, named job or prepared-file message is not a full external backup.

For a recovery rehearsal: export Complete, verify the JSON exists outside NightSweep, reopen it, confirm the job, route, power, records, practice progress and all six originals, then return to the protected real job. Keep independent backups before updates, uninstalling, clearing storage or moving among app/browser stores.

Offline use, updates and Help

Where: browser Home → NightSweep on this device; header/Home → Help. **Practice:** Complete for backup; Custom curves → Help; any loaded exercise for an offline calculation check.

Function	Use	Result and limit
Ready for offline use	Wait for the installed browser shell to report readiness	Confirms required cached shell availability; a visible online page alone is insufficient
Install app	Use the browser's offered prompt where supported	Optional Home Screen installation; full phone-specific steps are in the separate installation guide
Retry offline setup	Retry the reported failed shell preparation while online	Does not require clearing job storage
Check for updates	Look for a newer hosted release	Does not automatically replace the running app
Install update → Reload app	Export current work, apply the offered update, then reload when ready	User-controlled update sequence; check version and saved jobs afterward
Protect local storage	Request persistence where supported	Best effort, not a backup or protection from device/browser clearing/uninstall
Keep your work safe	Review local storage guidance/status	Helps identify the active store; no cloud sync or cross-browser sharing is added
Help task shortcuts	Choose the matching task from Help	Opens the existing live screen

Function	Use	Result and limit
Methods and limits	Read equations, assumptions, terminology and unavailable-data behavior	Explains interpretation; source links need connectivity
Run build self-check	Run bundled reference cases and read status	Checks those build cases; a pass is not physical device or plant certification. A failure warns against trusting results

Once ready, the installed browser app can reopen its cached shell and use bundled calculations/training offline. External manuals still need internet. The standalone HTML and native wrapper have different delivery/storage behavior; opening a local HTML file does not establish browser PWA installation or reliable local-file storage.

Detailed Android/iOS installation, compatible Preview update precautions and APK verification are documented in the separate installation guide.

Checked example answers

These are reference answers for the unchanged **synthetic** examples. Restore a baseline before comparing. They do not establish field acceptance or manufacturer settings.

Exercise/function	Inputs	Checked answer
Complete launch	54/860 MHz, raw 11/23 dBmV, remaining TP 20 dB	31/43 dBmV
Gain helper	54 MHz, input 7.1/output 16.1 dBmV, +20 dB each	Corrected 27.1/36.1 dBmV; net gain 9 dB
Complete A1 output	Entered route and 9/18 dB gain anchors	About 36.94/47.61 dBmV at 54/860 MHz
Complete A2 output	Entered route and 54/550/860 MHz gain table	About 40.55/45.62 dBmV at 54/860 MHz
Complete EOL Tap C	Entered mixed-brand route	About 14.01/4.94 dBmV at 54/860 MHz
Complete drop loss	75 ft custom RG-6, rates 1.8/6.2 dB per 100 ft plus custom 4 dB	Cable 1.35/4.65 dB; total 5.35/8.65 dB at 54/860 MHz
Complete TDR	A2 origin 1700 ft; displayed 295 ft; lead 25 ft; equal VF 0.87	Path 270 ft; coordinate 1970 ft; 160 ft into r5's 310 ft span
Complete frequency checker	860 MHz; 200 ft × 2.2 dB/100 ft + 1.4 dB; 43 dBmV launch	Loss 5.8 dB; receive 37.2 dBmV
Complete power	90 V / 15 A; synthetic 0.75/0.60/0.45 A loads; 20 W / PF 0.9 branch	2.047 A; 13.65% loading; minimum 88.83 V at A2

Exercise/function	Inputs	Checked answer
Complete power trial	Add 300 ft P3 .625 and 0.75 A after A2	2.798 A; 18.65%; 87.37 V at new load
Six serial AOI actives	Each incoming 300 ft custom span loses 3/9 dB; each entered net gain restores 3/9 dB	All six marker outputs 34/46 dBmV at 261/993 MHz; source node excluded from six-active count
Custom reverse TDR	100 ft displayed, actual VF 0.88/meter VF 0.80, lead 25 ft; reverse from 200 ft	Path 85 ft; route coordinate 115 ft
Voltage unit converter	72 dBµV	12 dBmV, same plane
Separate Records injection arithmetic	Desired 40 + remaining coupling 20 + remaining lead 1	61 dBmV manual-setting estimate; not automatically ONX Injection

The fresh independent complete-workflow audit compares actual application values with Wolfram references, including baseline/trial power and the explicit custom drop. Retained evidence lives with the release under `research/alpha18/`. Arithmetic agreement checks the supplied assumptions and inputs; it does not verify installed equipment, waveform, actual losses or phone behavior.

Station worksheet

Open **Route → Station**, choose the installed node or active and select **Before adjustments** or **After adjustments**. Each stage keeps its own forward pilots, targets, actual readings and settings. Company targets/tolerances start unknown. The same sourced solver used in Calculators provides GainMaker input-part advice, predicted remaining errors, limit warnings and the manufacturer procedure window; re-measure to confirm. Quantum18 uses its electronic procedure without plug-in part picks.

The return worksheet keeps reference-point target plus applicable net insertion loss, actual physical injection and same-plane receiver reading/reference separate. Its reminders follow the selected plant and exact equipment; manufacturer recommendations and the editable company TX entry never fill injection/receiver fields. Record **Balanced** and **Swept** separately with their own dates. These are technician records, not automatic acceptance.

Incomplete numeric or date edits are marked beside the field and do not replace the recorded entry. Finish the entry, or deliberately clear every part to record unknown. Leaving an incomplete field or returning to Route → Station restores its recorded value; a completely cleared native date is committed when focus leaves the control. Viewing an imported invalid value does not erase its stored text. Balance and sweep remain separate records.

Complete practice has **22 steps**, including the GainMaker station example; six-active practice has **16 steps**, including the separate electronic station example. Back up and restore the worksheets with the whole job. Older worksheets with incomplete binding metadata stay visible/read-only until explicit station confirmation; the original is retained in the backup. Catalog band corrections do not detach saved observations.

Night log and shift handoff

Where: Job → Night log. **Practice:** Six active devices in series → Five-station night handoff.

The **Unknown** source tag beside **Recorded entries** documents the existing **200 entries/job** application guard. The same row appears in **Test → Limits** under **Night log application guards**. This software storage/display bound is not an RF limit, equipment capability or company acceptance criterion; device storage can fill sooner.

Keep personal ticket/change, node/amp label, actions, ROC/dispatch updates, gas/fuel, mileage, as-built discrepancies and next-shift/open-issues notes with this job. Keep customer names, addresses and account numbers out. Window start/end are optional typed `YYYY-MM-DDTHH:mm` device-local wall times; the zone is not recorded and blanks stay unknown. **Added** is the device timestamp when you press **Add entry**, not when the field work necessarily occurred. Enter fuel and mileage as text with your own amount, reading/distance and units; no conversion, total or reimbursement rule is assumed.

Incomplete or invalid window text shows a named format message beside that field and is never saved over the previous value. Leaving the field or returning to Night log restores its saved value (blank means unknown) and explains the restoration beside the field. Enter a complete valid time to replace it, or clear the entire field deliberately to record unknown.

The selector starts at **General update / no Station snapshot**. To capture readings, first save the worksheet under **Route → Station**, then explicitly select it in Night log and press **Add entry**. Both actual Before and After stages, raw inputs, measurement planes, frozen results and separate balance/sweep records are copied once. You can edit entry notes; later Station edits never refresh the capture. Add another entry for a new capture. Changed/removed equipment is labeled historical; read-only and unconfirmed legacy worksheets retain their warnings and old dates. Unknown targets, readings, planes or results remain unknown with reasons. Recorded work does not establish RF acceptance.

Generate handoff creates a selectable full-text preview above the entry editors. Editing saved log content clears that preview and asks you to generate it again; exports always use the current saved log. **Export TXT** and **Export CSV** prepare files through the existing offline save/share path; finish the dialog and check the external file. CSV has one rectangular header, separate Before/After rows per captured entry and a blank-stage row for a general update. It retains raw values/planes/context and independent work dates alongside formatted values. Levels and derived deltas show two decimals, pilot MHz up to three; rounded zero has no sign. Formula-like text receives the existing spreadsheet-safe apostrophe prefix in CSV. Text exports exclude legacy address/lead/project-note fields and attachments.

Files → Back up project remains the full-fidelity portable backup: exact saved text, unrounded captured results and original attachments. Restore the JSON to recover the log and both stages. For practice, manually record invented A1–A5 Station worksheets in the six-active exercise, capture each explicitly, generate the handoff and reopen TXT/CSV. The short five-station example fits the defined phone preview; longer notes can scroll without truncation. Practice exports keep the synthetic warning.

Sweep due

Open **Job** → **Sweep due** for personal reminders from saved Station work dates across this device's jobs, including the current working job. It shows the installed launch node/active and route amplifiers, with separate sweep and balance dates, date validity and completed calendar-month ages. A missing worksheet stays visible as unknown. A uniquely matching saved job ID is overlaid with current unsaved dates; duplicates have independent job histories. Ambiguous IDs and unreadable sources produce visible diagnostics.

Control / result	Meaning
Sweep reminder interval (whole calendar months)	Starts unknown. Enter digits only, 1–1200 months; the range is an Unknown app input guard. Raw invalid text remains visible.
Saved preference	Saved with the current working job; used for this view of all jobs. Loading a different job can change it. Autosave, named duplicate and full JSON backup preserve this optional preference.
Refresh list	Read actual Station/library changes and today's local device date. Entering/resuming the visible tab also refreshes. No background monitoring.
Before reminder window / Reminder window / Due / overdue	Green / yellow / red reminders, with words and icons; never RF acceptance. Practice/sample counts stay separate from work reminders.
Chosen sweep due date	Original recorded sweep date plus entered calendar months, clamped to the target month's final day; no 30-day-month arithmetic.
Last recorded balance	Independent technician event/date/age. It does not reset sweep age or receive a balance cadence.
Previous / Next	Pages of at most 25 entries; the full source and station counts remain visible.

Yellow starts at `ceil(2 × interval / 3)` calendar months, tagged **Unknown** as an unverified application policy implementing the build-list 12-of-18-month example. Due / overdue wins when warning and due dates coincide. An older plan mentions 18 months; its Hranac attribution remains unverified. Confirm your company's interval. This is a personal reminder setting, not a company specification supplied by the app.

Dates use exact Gregorian `YYYY-MM-DD`, years 0001–9999. Future, invalid or unavailable dates remain neutral. Only a currently bound, writable Station record with Swept recorded and a valid nonfuture sweep date supplies a current reminder. Unchecked-with-date and recorded-without-date states stay distinct. Binding-unconfirmed historical and read-only UBT worksheets preserve dates, completed ages, validity and done facts with a historical reason, without current-service reminder wording or colors. Binding confirmation retains the original actual event date. Metadata, training progress, captured history and calculator results never supply work dates.

The interval belongs to the open job, including in an empty list. **Not saved** feedback appears directly below the interval if autosave fails; export a project backup before closing. Feedback belongs only to that job and raw interval.

Edit actual dates in **Route → Station** and manage jobs in **Job → Details**; this view does not open a saved job or edit work dates. The six-active **Personal sweep reminders** step explicitly enters 18 for demonstration and clears it again. At explicit 2026-11-11 in the focused QA fixture, three named synthetic jobs swept 0/12/19 months before today show green/yellow/red. Practice provenance remains visible after edits, copies and JSON restore.

53-feature coverage index

This index follows the feature choices in the app. Choose the same label in Home's feature index to load the matching seeded practice. Sections above also cover controls that share one indexed feature, such as export buttons and station history.

#	Feature choice	Reference section	Loadable exercise
1	Home, practice and backup	Header and Home	Complete
2	Help, methods and build self-check	Offline and Help	Custom curves
3	Launch, source, correction and temperature	RF setup	Complete
4	Route, branches and distances	Cable route	Complete
5	Amplifier gain and reserve	Gain	Complete
6	Installed equipment and exact manuals	Equipment setup	Complete
7	Equipment catalog and source references	Catalog	Complete
8	Forward and return setup	Sweep setup	Complete
9	Forward TDR and active boundaries	TDR	Complete
10	Absolute, normalized and loss charts	Predicted sweep	Complete
11	Single-path frequency calculator	Frequency	Complete
12	Meter readings and prediction error	Readings	Complete
13	Cable, tap and target references	References	Complete
14	Power load, voltage and branches	Power	Complete
15	Power what-if and Keep	What-if	Complete
16	Named jobs and work-order details	Job details	Complete
17	Original files and supported CSV	Evidence	Complete
18	Leakage, maintenance and conversion	Checklists	Complete
19	Forward, return, punch and review	Records	Complete

#	Feature choice	Reference section	Loadable exercise
20	Station balance, return worksheet and separate sweep dates	Station worksheet	Complete and six-active (2.0 Alpha build 3)
21	Run sheet, share, print and CSV exports	Exports	Complete
22	Six serial active devices	Practice library, checked answers	Six actives
23	Entered amplifier gain table	Gain	Six actives
24	Derive gain from two corrected readings	Gain helper	Complete
25	Tap faces and RG-6 drop splitter	Point tools	Complete; explicit custom loss
26	Map note and step-by-step arithmetic	Point tools	Custom curves
27	Custom cable and passive loss tables	Cable route	Custom curves
28	Manual source and launch table	RF setup	Custom curves
29	Reverse TDR and velocity correction	TDR	Custom curves
30	Available carriers and local targets	References	DSIM
31	OEM AGC installed control path	Equipment controls	OEM AGC
32	DSIM and historical pilot-code lookup	Equipment controls	DSIM
33	Manual / thermal installed control path	Equipment controls	Manual / thermal
34	Conventional GS7K procedure	Equipment controls	GS7K node
35	dBµV ↔ dBmV conversion	Unit conversion	Complete
36	Return injection and receiver records	Records, sweep setup	Complete
37	Before / after evidence comparison	Evidence	Complete
38	Portable backup and recovery	Full backups	Complete
39	Field calculators and source tags	Field calculators	Custom curves (2.0 Alpha build 3)
40	Limits, labels and sources	Limits and sources	Custom curves (2.0 Alpha build 3)
41	DOCSIS health card	DOCSIS health card	Custom curves (2.0 Alpha build 3)
42	Leakage limits, CLI and leak log	Leakage	Custom curves (2.0 Alpha build 3)
43	Outage first-response prompts	Outage first response	Custom curves (2.0 Alpha build 3)
44	Night log and shift handoff	Night log and shift handoff	Six active devices in series (2.0 Alpha build 3)

#	Feature choice	Reference section	Loadable exercise
45	Personal sweep reminders and independent balance dates	Sweep due	Six actives (2.0 Alpha build 3)
46	Attached sweep grading aid	Sweep grading aid	Custom curves (2.0 Alpha build 3)
47	Impairment cards and standing-wave search	Impairments	Custom curves (2.0 Alpha build 3)
48	Fiber color finder and link loss budget	Fiber	Custom curves (2.0 Alpha build 3)
49	High split band plans, aeronautical band and Remote PHY bring-up	High split	Custom curves (2.0 Alpha build 3)
50	Why cards: the SCTE 293 basics	Why cards	Custom curves (2.0 Alpha build 3)
51	FCC proof-of-performance limits and test rules	Proof of performance	Custom curves (2.0 Alpha build 3)
52	Study mode: practice questions in four domains	Study mode	Custom curves (2.0 Alpha build 3)
53	Supply PM notebook and span voltage drop	Supply PM	Custom curves (2.0 Alpha build 3)

Build 3 attached sweep grading aid

In **Job → ONX files → Grading aid**, select an attached **Sweep**, then explicitly choose **42 / 54** or **204 / 258 MHz** and **Forward** or **Reverse**. Each of the four company profiles starts entirely unknown. Selecting a profile does not edit an old job. Enter measured marker frequencies, a PTV limit, absolute level targets and tolerance, and a signed tilt target and tolerance as needed. Zero is an entered value; blank stays unknown. Both markers must exist exactly in the supplied samples; inspect the original preview for available frequencies. No interpolation or extrapolation occurs.

Supported native previews cover the inspected VIAVI **TestGroup** and **CDM2** sweep JSON layouts, directly or inside gzip (including `.json-1.gz`) or a ZIP with one eligible JSON. Only one unambiguous explicit trace is decoded; unsupported or ambiguous files stay attached with a reason. These narrow adapters are not a general VIAVI decoder. Native direction and reference state are shown separately: unknown direction remains unknown, even when you explicitly select a grading direction. A known native direction conflict blocks grading. Native limits never become company criteria. The included interchange CSV practice traces are invented and are not native ONX captures.

Grade selected sweep reads the saved original again and checks its size and SHA-256 before parsing. The result shows raw PTV (maximum minus minimum), low/high values, signed high-minus-low tilt, and each comparison's pass/fail/unknown reason. Raw absolute measurements use **dBmV**; differences, PTV and tilt use **dB**. Relative dB response cannot establish absolute dBmV levels. An empty referenced trace does not become a zero line or a flatness pass. Unreferenced absolute PTV is not normalized ripple.

A criterion that cannot be saved stays in this session with **Not saved** beside the named field and an autosave-failure status. Use **Back up project** to keep these changes outside the app; the warning clears after a successful whole-job autosave. Invalid text keeps its named error and is refused by **Grade selected sweep**,

including repeated taps. Correct it or deliberately clear it for unknown. If the panel refreshes, it restores the last valid in-session criterion and clears the invalid draft; an unsaved-value warning remains until saving succeeds.

Read-only numbers use up to 12 significant digits for readability. **≈0** identifies a computed difference within the existing machine-roundoff allowance; it adds no measurement tolerance. Editable criteria, stored raw numbers, exported history and pass/fail decisions are unchanged by display formatting. Very small entered targets and limits remain nonzero, using scientific notation when needed.

No TPC/test-point addback, lead correction, reference subtraction, normalization or detrending is applied. Verify that the selected targets and original share the same measurement plane. Application entry/decoding bounds are software resource guards, not OEM specifications. A grading result is your technician aid, never official acceptance; it does not set balanced/swept status or a sweep date.

Each compact historical snapshot saves its criteria, plant/direction, original evidence identity and SHA, job/source/record context and calculation time. New snapshots retain a SHA-256 digest of the relevant bound Station identities and saved records, including binding version, both Before/After worksheets and separate balance/sweep records. Archived bindings stay in the job backup and do not enlarge grading history. Legacy-unconfirmed worksheets remain unconfirmed. Earlier raw-context grading snapshots open and round-trip unchanged; they remain historical and require a new grade for the current context. Measurement time stays unknown unless you enter a verified time; calculation time is not the meter's measurement time. Changes make the current display stale/unknown. Old snapshots keep their original labels and values. Removing an attachment preserves its grading history with an unavailable-original message. A full **Back up project** includes the unchanged compressed/archive original and grading snapshots; restore remaps each evidence identity, including duplicate-byte attachments. Browser autosave and named-job duplication also preserve history. Keep external backups before clearing app data.

The **Custom curves and reverse TDR** exercise adds an explicit grading step and a feature-index link. It supplies instructions for a three-point invented interchange CSV (270/390/510 MHz, 0/2.5/0 dBmV). Enter the illustrative 3 dB PTV limit yourself: PTV is 2.5 dB and the other blank criteria stay unknown. These practice values are not shipped company defaults.

Known limits and unresolved information

Exact installed **GainMaker GaN control/commissioning**, **Quantum18 LE/BA I&O**, **EN9000/RFAM/RPD commissioning**, and some **DSIM-A/module-specific** procedures remain documentation gaps. Preserve them as unresolved; another model's manual is not a substitute. Original manuals, actual firmware/options, approved pilots, physical measurement planes, actual available carriers and operator criteria require job-specific confirmation.

NightSweep does not connect to, program or configure ONX, nodes, amplifiers, DSIM or XPERTrak. Unsupported meter formats remain originals; no OCR or automatic reading extraction is added. No source table, synthetic file, checklist, software status or reference calculation establishes installed RF acceptance.

Automated checks, independent arithmetic and desktop mobile-viewport testing do not establish physical Android/iPhone installation, WebView/Safari retention, native save/share/print behavior, touch/rotation, storage eviction or actual quota exhaustion. Preserve external backups and separately verify the phone and plant used for field work.